Towards Prediction of Optimal Camera Exposure Parameters for Image Classification

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1 Introduction

Recent work [1] has demonstrated that auto-exposure (AE) settings of a camera are highly suboptimal when the task is to maximize the classification performance: While AE settings often result in visually pleasing images, common classifiers yield significantly better predictions, when ISO, shutter speed, and aperture are adapted for the downstream classification network. Alongside this observation, [1] introduces both the *ImageNet-ES Diverse* dataset, in which 1,000 ImageNet [4] images are recorded under six lighting directions with three ISO, shutter speed, and aperture values each (27 exposure variants per lighting condition), and *Lens*, which captures several settings and keeps the prediction with the highest softmax value. Yet, an exhaustive search over such parameters requires a large number of acquisitions and therefore induces high latency dependent on the camera sensor [1], making it unsuitable for highly dynamic scenes or video capture with varying environmental conditions. While the authors studied different candidate selection algorithms to reduce the number of captures from 27 down to 18 or 6, their primary choices are based on random or grid-based sampling, as well as on lowest capture time. Although this does not require all 27 variants to be recorded, in the best-case scenario it still results in an approximately sixfold increase in latency.

In this extended abstract, we propose to *learn the direct prediction of the optimal camera exposure parameters from a single auto-exposure image*, see Fig. 1. Note that while the prediction itself requires only this one image, acquiring the optimized capture the classifier operates on adds a second exposure, hence our two-capture setting. We use a lightweight parameter prediction network, study different features, representations, and losses for camera parameter prediction and demonstrate that our proposed two-capture setting surpasses, on average, all Lens candidate selection algorithms from [1], with our best variant yielding a classification accuracy only 1.4 pp short of the full (exhaustive) classification of 27 different captures proposed in [1]. While our investigation is not yet exhaustive, we believe the observed effects are promising, and we present this extended abstract to discuss our early results and open questions with the community.

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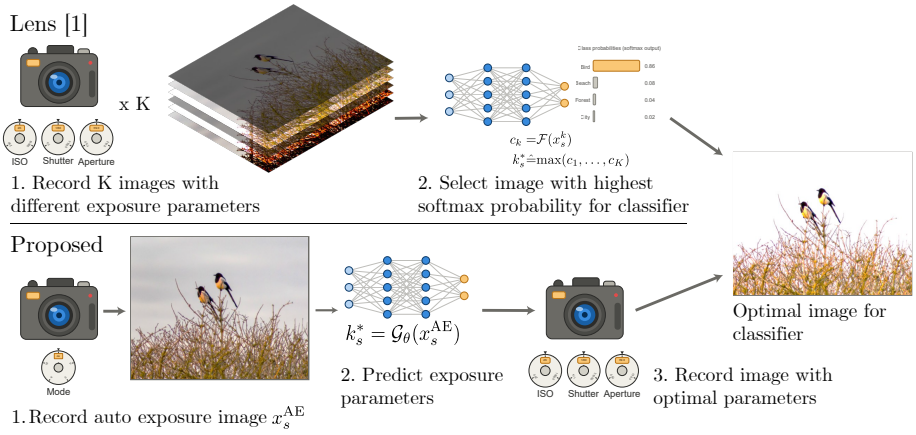


Fig. 1. Illustration of our proposed pipeline (bottom) in comparison to prior work (top): Instead of directly taking multiple captures, we propose to train a network $\mathcal{G}_\theta$ on predicting the optimal camera parameters for a given classifier from an auto-exposure image x_s^{AE} in a single shot.

2 Related Work

Classical AE methods use image statistics to produce visually pleasing exposures [7,8]. Task-aware alternatives optimize acquisition for a downstream objective. Onzon et al. jointly learn exposure selection, image processing, and object detection for high-dynamic-range scenes [11]; feature-space exposure fusion uses several captures [10]. CamTuner employs reinforcement learning and a virtual camera to adapt non-automatic parameters for deployed video analytics [12]. ImageNet-ES demonstrated that physical environment and sensor shifts differ from standard digital corruptions and that sensor control can improve robustness without changing the recognition model [2]. The follow-up work [1] proposes the *Lens* method discussed above and is the approach we aim to improve upon.

3 Methodology

For a scene s, let $x_s^{\mathrm{AE}} \in \mathbb{R}^{3\times H\times W}$ be the AE image with image dimensions H, W, and let $\mathcal{P} = \{p_1, \ldots, p_N\}$ be the available settings of the camera. Using the ImageNet-ES Diverse dataset [1], we consider $N = 27$ combinations of three ISO, three shutter-speed, and three aperture values. During training, the scene is also captured as $x_{s,k} \in \mathbb{R}^{3\times H\times W}$ under every $k \in \{1, \ldots, N\}$, i.e., for every setting p_k. For a fixed frozen classifier $\mathcal{N}$ (with softmax probabilities as its output), our goal is to train a policy $\mathcal{G}_\theta$ such that ideally

$$\mathcal{G}_\theta(x_s^{\mathrm{AE}}) = k_s^* \qquad \text{for} \qquad k_s^* = \arg\max_k (\mathcal{N}(x_{s,k}))_{y_s}, \tag{1}$$

where y_s denotes the true label of scene s. Note that we target the setting maximizing $\mathcal{N}$'s belief in the *true* label, rather than its overall confidence, which

may be high for a wrong prediction. To construct such a parameter prediction model $\mathcal{G}_\theta$, we extract k_s^* for all images in the training dataset and investigate two different architectures, yielding three trained variants:

1. **Lighting aware architecture.** We feed the six-dimensional one-hot encoded information about the lighting direction provided in the ImageNet-ES Diverse dataset into a small 2-layer MLP to obtain an intermediate representation $f_{\text{light}} \in \mathbb{R}^{64}$ which is concatenated with an intermediate feature dimension from a pretrained ResNet-18 [5] image classifier $\mathcal{F}(x_s^{\text{AE}}) : \mathbb{R}^{3\times H\times W} \rightarrow \mathbb{R}^{512}$ on the AE image. The concatenated features are fed into a 2-layer MLP with ReLU activation and 30% dropout on the first layer. The policy network $\mathcal{G}_\theta$ is then trained against a one-hot encoding of the best-performing setting k_s^* using a class-balanced cross-entropy loss and label smoothing [9] with $\alpha = 0.05$ which we denote as *LACE* approach. Additionally we train the same architecture using a Hinge loss [3] referred to as *LAGE*, to facilitate that the optimal parameter setting k_s^* exceeds all other settings by a pre-defined margin $m = 1$.
2. **Histogram-based (HB) architecture.** As the lighting information might not be universally available, we also experiment with a 2-layer MLP architecture, which uses the features of $\mathcal{F}(x_s^{\text{AE}})$ concatenated with a patch-wise image histogram[1] $f_{\text{hist}} \in \mathbb{R}^{74}$ as input. The MLP is then trained using a Huber loss [6].

4 Preliminary Results

We evaluate our approach using the ImageNet-ES Diverse dataset [1] with a ResNet-18 classifier for $\mathcal{N}$ and ImageNet-pretrained weights [4]. To ensure a fair comparison across all methodologies, we isolate a held-out test split of 600 unique physical scenes (including lighting variations yielding a total of 16,200 test images). Further 600 unique physical scenes are used for validation and the remaining 4800 physical scenes (129,600 images) are used for training.

We compare against the naive *AE baseline*, i.e., directly evaluating $\mathcal{N}(x_s^{\text{AE}})$; *Lens Full* [1], selecting the setting with the highest softmax probability among all $N = 27$ recordings, along with its candidate selection algorithms CSA1–3; and the *theoretical Oracle-S*, the upper bound reachable by $\mathcal{N}$ using k_s^* from (1) (requiring ground truth).

Table 1 presents our numerical results. All our three implemented variants surpass the AE baseline as well as the discussed candidate selection techniques (CSA1–3), which we re-implemented based on the descriptions in [1]. Further, our frozen classifier reaches the stated accuracies by capturing only two images, cutting image acquisition costs by 2/3 and increasing accuracy by up to 5.81 pp over the best six-capture candidate selection CSA2 [1] (1.88 pp when compared

[1] This statistics feature branch consists of 60 histogram (4 patches, 3 channels, 5 bins), 9 global (per-channel mean, standard deviation and clipping ratios) and 5 spatial luminance (patch-wise grayscale mean and global variance) features.

against the CSA2 value reported in [1]). At the same time, policy network inference takes only 3.5 ms (LACE, LAGE) and 5.15 ms (HB)[2]. Accuracies of Lens in Table 1 are our reproduced results, while values reported in [1] are given in brackets[3]. We observe a gain when using light metadata, yet to this date we could not verify whether this is due to larger network sizes based on different dimensionalities or actual information usage. A substantial gap of roughly 18 pp remains between our best variant and the oracle upper bound (Table 1) – either due to incomplete convergence or the inability of the proposed architectures to extract generalizable information from the AE image for accurate classification. This indicates considerable headroom for better information utilization and architectural choices, which we aim to explore in future work.

Table 1. Accuracy of a ResNet-18 on the ImageNet-ES Diverse [1] classification dataset under different camera parameter prediction methods: While all approaches can significantly improve over the poor performance of the naive AE, our proposed approach surpasses the 6-capture CSAs of [1] with only two required captures. Accuracies are averaged over three runs (± std. where applicable); original values from [1] in brackets.

Evaluation Pipeline	**Accuracy (%)**	**Number of Captures**
Naive AE	12.17	1
LACE (Ours)	28.39 ± 0.35	2
LAGE (Ours)	29.28 ± 0.25	2
HB (Ours)	24.22 ± 1.65	2
Lens CSA1 (Random) [1]	23.33 ± 1.27 (26.9)	6
Lens CSA2 (Grid Random) [1]	23.47 ± 1.23 (27.4)	6
Lens CSA3 (Fast Acquisition) [1]	21.83 (25.1)	18
Lens Full [1]	30.67 (34.6)	27
Theoretical Oracle-S (Upper Bound)	47.67 (52.6)	27

5 Conclusion and Future Work

This extended abstract demonstrated the feasibility to train a neural network on directly predicting discrete ISO, aperture and shutter speed values from a single auto-exposure image in such a way that a subsequent classifier benefits from the recorded setting. Our approach surpasses, on average, the candidate selection approaches of Lens in [1] while requiring only two (instead of six) recordings in our preliminary experiments. In future work, we plan to evaluate all classifiers used in [1], and will provide a detailed ablation study on the architectures, inputs, and losses used for camera parameter prediction.

[2] Measured on a NVIDIA T4 GPU for a single AE image.

[3] Since neither code nor checkpoints are available, we use our own pre-trained ResNet-18 $\mathcal{N}$, which likely explains the numerical differences to [1].

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